

Jeremy L Thompson

Research Software Engineer

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English (native), German (A2-B1)

Education

- 2021 **PhD**, *University of Colorado Boulder, Applied Mathematics*
- 2012 **MSc**, *University of Washington, Applied Mathematics*
- 2009 **BS**, *United States Air Force Academy, Mathematics, Minor in Philosophy*

Experience

- 2026 - **Research Software Engineer**, *Material Systems Modeling, Helmholtz-Zentrum Hereon*
 - Developer for 4C multiphysics simulation framework - github.com/4C-multiphysics/4C
 - Leading porting of CPU particle physics module to GPU computation via Kokkos
- 2025 - 2026 **Research Assistant Professor**, *CS Department, University of Colorado Boulder*
- 2021 - 2025 **Research Software Engineer**, *CS Department, University of Colorado Boulder*
 - Architect for solid mechanics library with PETSc and libCEED - gitlab.com/micromorph/Ratel
 - Lead developer for performance portable finite element library - github.com/CEED/libCEED
 - Maintainer for fluid dynamics library with PETSc and libCEED - gitlab.com/phypid/HONEE
 - Mentor graduate students; teach software development and academic research best practices
 - Quality focused; testing, maintainability, and documentation while expanding functionality
 - Research software innovations; developed GPU matrix-free Material Point Method software
 - Taught: Numerical Computation
- 2017 - 2021 **Graduate Research Assistant**, *Applied Math Department, University of Colorado Boulder*
 - libCEED core developer - C99 library with CPU/GPU performance portability; AVX, CUDA, HIP, & SYCL impl; C/C++, Fortran, Rust, Julia, & Python interfaces - github.com/CEED/libCEED
 - Architect/developer for FEM preconditioner analysis toolkit - github.com/jeremytl/LFAToolkit.jl
 - Developed efficient implementations of high order finite elements for new exascale hardware
 - Researched Local Fourier Analysis for parameter tuning with sharp convergence estimates of preconditioners for arbitrary order FEM based operators, including p-multigrid and BDDC
- 2014 - 2016 **Assistant Professor**, *Math Department, United States Air Force Academy*
- 2012 - 2014 **Instructor**, *Math Department, United States Air Force Academy*
 - Math majors coordinator; ensured 50+ students in majors on track, organized majors events
 - Research mentor; advised students for independent research in math and operations projects
 - Faculty club advisor, Cadet Honor Guard and Freethinkers club; mentored student leaders, coordinated club travel and budget, supervised chemical and explosive safety programs
 - Taught: Calc I/II/III, Differential Equations, Engineering Mathematics, Discrete Mathematics
- Summer 2014 **Visiting Scientist**, *Lawrence Livermore National Laboratory*
 - Improved wind forecasting data projections for optimizing power grid production balancing
 - Investigated and compared smoothing filters, FFT, Gaussian smoothing, and non-local means
- 2009 - 2012 **Advanced Weapon Systems Analyst**, *United States Air Force*
 - NUCWSEP tester and analyst; conducted live tests of B-52 Air Launched Cruise Missile
 - Aggregated and analyzed ACC aircraft nuclear weapon test results for annual planning report
 - Overhauled annual ALCM accuracy and reliability forecasts; restored USSTRATCOM confidence

Honors and Awards

- 2020 - 2025 **Annual freeCodeCamp Top Contributor Award**, *freeCodeCamp*
- 2018 **Helping Hands Volunteer Award**, *Moving to End Sexual Assault*
- 2016 **Brigadier General Daniel W Litwhiler Award for Outstanding Course Director**, *USFA Department of Mathematical Sciences*
- 2014 **Outstanding Academy Educator**, *USFA Department of Mathematical Sciences*
- 2013 **Outstanding New Instructor**, *USFA Department of Mathematical Sciences*
- 2011 **Junior Military Scientist of the Year**, *USAF Air Combat Command*
- 2010 **Honor Graduate**, *Operations Research Systems Analysis Military Application Course*
- 2008 **Excellence in Student Exposition and Research**, *American Mathematical Society*

Selected Publications

ORCID: orcid.org/0000-0003-2980-0899
ResearchGate: researchgate.net/profile/Jeremy-Thompson
Google Scholar: scholar.google.com/citations?user=UCKh6wcAAAAJ

- [1] Zachary R. Atkins, Jed Brown, Fabio Di Gioacchino, Layla Ghaffari, Zachariah T. Irwin, Rezgar Shakeri, Karen Stengel, and Jeremy L. Thompson. Ratel: Performance portable solid mechanics with libCEED and PETSc. *Journal of Open Source Software*, 11(118):8388, 2026.
- [2] Zachary R. Atkins, Jed Brown, Rezgar Shakeri, and Jeremy L. Thompson. Matrix-free finite element methods with arbitrary quadrature point locations. *Computational Science and Engineering*, 3(1):3, 2026.
- [3] Jed Brown, Ahmad Abdelfattah, Valeria Barra, Natalie Beams, Jean-Sylvain Camier, Veselin Dobrev, Yohann Dudouit, Leila Ghaffari, Tzanio Kolev, David Medina, Will Pazner, Thilina Ratnayaka, Jeremy Thompson, and Stan Tomov. libceed: Fast algebra for high-order element-based discretizations. *Journal of Open Source Software*, 6(63):2945, 2021.
- [4] Jed Brown, Valeria Barra, Natalie Beams, Leila Ghaffari, Matthew Knepley, William Moses, Rezgar Shakeri, Karen Stengel, Jeremy L. Thompson, and Junchao Zhang. Performance portable solid mechanics via matrix-free p -multigrid, 2022.
- [5] Rachel Eaton, Kurt Herzinger, Ian Pierce, and Jeremy Thompson. Numerical semigroups and the game of sylver coinage. *The American Mathematical Monthly*, 127(8):706–715, 2020.
- [6] Tzanio Kolev, Paul Fischer, Misun Min, Jack Dongarra, Jed Brown, Veselin Dobrev, Tim Warburton, Stanimire Tomov, Mark Shephard, Ahmad Abdelfattah, Valeria Barra, Natalie Beams, Jean-Sylvain Camier, Noel Chalmers, Yohann Dudouit, Ali Karakus, Ian Karlin, Stefan Kerkemeier, Yu-Hsiang Lan, and Vladimir Tomov. Efficient exascale discretizations: High-order finite element methods. *The International Journal of High Performance Computing Applications*, 06 2021.
- [7] Rezgar Shakeri, Leila Ghaffari, Jeremy Thompson, and Jed Brown. Stable numerics for finite-strain elasticity. *International Journal for Numerical Methods in Engineering*, page e7563, 2024.
- [8] Jeremy L Thompson. An emperical evaluation of denoising techniques for streaming data. Technical Report LLNL-TR-659435, Lawrence Livermore National Laboratory, August 2014.
- [9] Jeremy L. Thompson, Jed Brown, and Yunhui He. Local fourier analysis of p -multigrid for high-order finite element operators. *SIAM Journal on Scientific Computing*, 45(3):S351–S370, 2023.
- [10] Jeremy L Thompson, Kurt Herzinger, and Trae Holcomb. The frobenius number of balanced numerical semigroups. *Semigroup Forum*, 94:632–649, 2017.